📍 Cartographer Relocalization and Pure Localization Autonomous Navigation

1. Course Content

- Master using Google Cartographer to load previously built pose state file (`.pbstream`) for high-precision global pure localization (Pure Localization).
- Combine Cartographer pure localization pose output with Nav2 path planner to achieve autonomous obstacle-avoiding single-point/multi-point navigation in known map environments.

---

2. Cartographer Pure Localization Navigation Principle Analysis

In actual industrial and research applications, after map building is complete, there is usually no need to continue incrementally expanding the map during navigation. Cartographer pure localization mode has the following characteristics and advantages:

1. **Pose Graph Freeze (Freeze State)**:
   After loading existing `.pbstream` offline pose graph, Cartographer sets all historical submaps to read-only frozen state, no longer inserting new pose nodes and constraint edges, greatly reducing memory usage and CPU computational load.
2. **Real-time CSM Scan Matching and Kidnap Resistance**:
   The front-end uses Correlative Scan Matching to quickly perform graph matching between real-time laser scans and historical submaps; even if the robot's starting position deviates or it is short-distance transported (robot kidnapping), the algorithm can quickly re-converge localization based on surrounding geometric features at millisecond level.
3. **Seamless Nav2 Navigation Protocol Stack Integration**:
   Cartographer outputs authoritative `map -> odom` coordinate transformation, while Nav2 focuses on costmap (Costmap 2D) updates, A* global path planning, and local obstacle avoidance control.

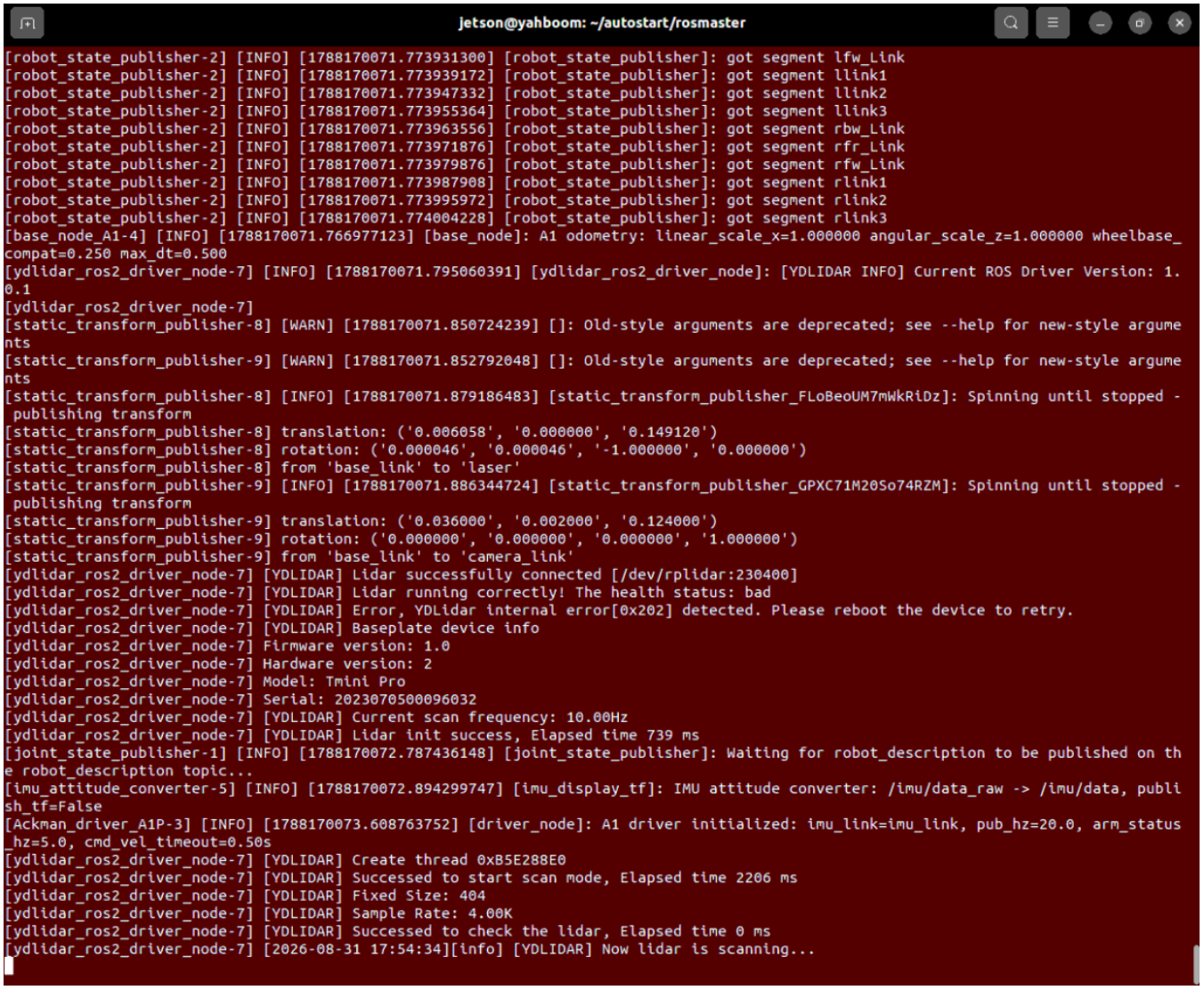
---

3. Preparation and Running

3.1 Start Chassis + Lidar

Run in **Terminal 1** (one-click start of A1 Pro chassis driver, IMU attitude solution, and lidar):

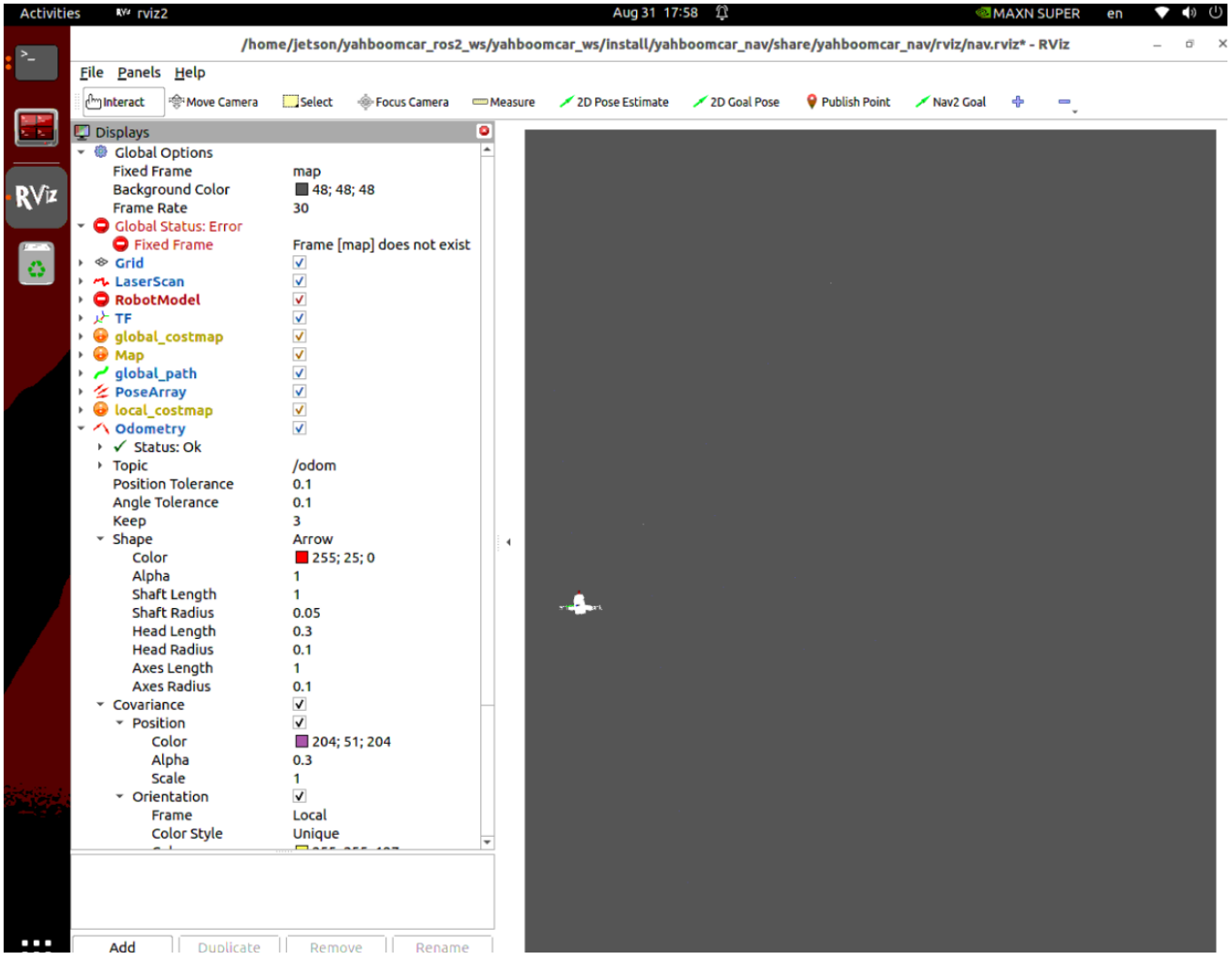
```
ros2 launch yahboomcar_nav laser_bringup_launch.py
```



3.2 Start RViz Navigation Visualization

Run in **Terminal 2** (it is recommended to open RViz on the virtual machine if the VM and robot are on the same LAN):

```
ros2 launch yahboomcar_nav display_nav_launch.py
```



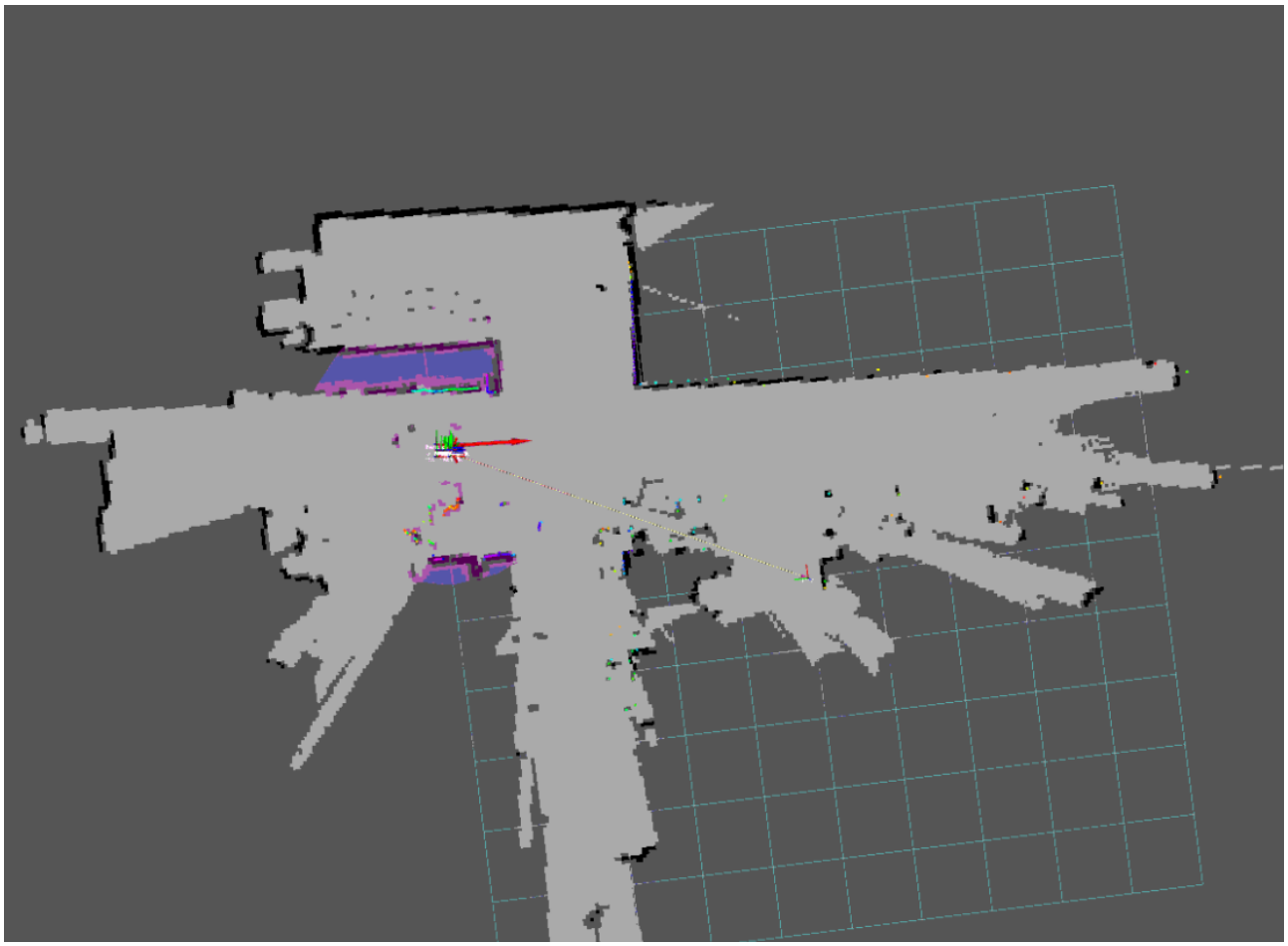
3.3 Start Cartographer Pure Localization Node

Run in **Terminal 3** to load `.pbstream` pose graph and start pure localization core:

```
ros2 launch yahboomcar_nav localization_imu_odom.launch.py use_rviz:=false
load_state_filename:=$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/map.pbst
ream
```

```
jetson@yahboom:~$ ros2 launch yahboomcar_nav localization_imu_odom.launch.py use_rviz:=false load_state_filename:=/home/jetson/map/map.pbstream
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/pose_graph.luapose_graph.lua'.
[cartographer_node-1] [INFO] [1788170604.259213680] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/pose_graph.luapose_graph.lua'.
[cartographer_node-1] [INFO] [1788170604.259476990] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder.luatrajectory_builder.lua'.
[cartographer_node-1] [INFO] [1788170604.259519520] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder.luatrajectory_builder.lua'.
[cartographer_node-1] [INFO] [1788170604.259595909] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder_2d.luatrajectory_builder_2d.lua'.
[cartographer_node-1] [INFO] [1788170604.259629734] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder_2d.luatrajectory_builder_2d.lua'.
[cartographer_node-1] [INFO] [1788170604.259807408] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder_3d.luatrajectory_builder_3d.lua'.
[cartographer_node-1] [INFO] [1788170604.259843922] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder_3d.luatrajectory_builder_3d.lua'.
[cartographer_node-1] [INFO] [1788170604.311704267] [cartographer logger]: I0831 18:03:24.000000 70766 map_builder_bridge.cpp:117] Loading saved state '/home/jetson
/map/map.pbstream'...
[cartographer_node-1] [INFO] [1788170604.518953543] [cartographer logger]: I0831 18:03:24.000000 70766 map_builder_bridge.cpp:136] Added trajectory with ID '1'.
[cartographer_node-1] [INFO] [1788170604.565510241] [cartographer logger]: I0831 18:03:24.000000 70766 ordered_multi_queue.cc:172] All sensor data for trajectory 1
is available starting at '639237674045638968'.
[cartographer_node-1] [INFO] [1788170604.594751697] [cartographer logger]: I0831 18:03:24.000000 70766 local_trajectory_builder_2d.cc:312] Extrapolator not yet init
ialized.
[cartographer_node-1] [INFO] [1788170604.645255745] [cartographer logger]: I0831 18:03:24.000000 70766 local_trajectory_builder_2d.cc:124] Extrapolator not yet init
ialized.
[cartographer_node-1] [INFO] [1788170604.707386407] [cartographer logger]: I0831 18:03:24.000000 70766 local_trajectory_builder_2d.cc:135] Extrapolator is still ini
tializing.
[cartographer_node-1] [INFO] [1788170604.795973862] [cartographer logger]: I0831 18:03:24.000000 70766 pose_graph_2d.cc:148] Inserted submap (1, 0).
```

> **Tip**
>
> After pure localization starts, the robot will automatically match the surrounding environment for alignment; if in a feature-sparse area, you can also use the `2D Pose Estimate` tool in RViz top toolbar to click and drag the heading arrow to assist acceleration convergence:

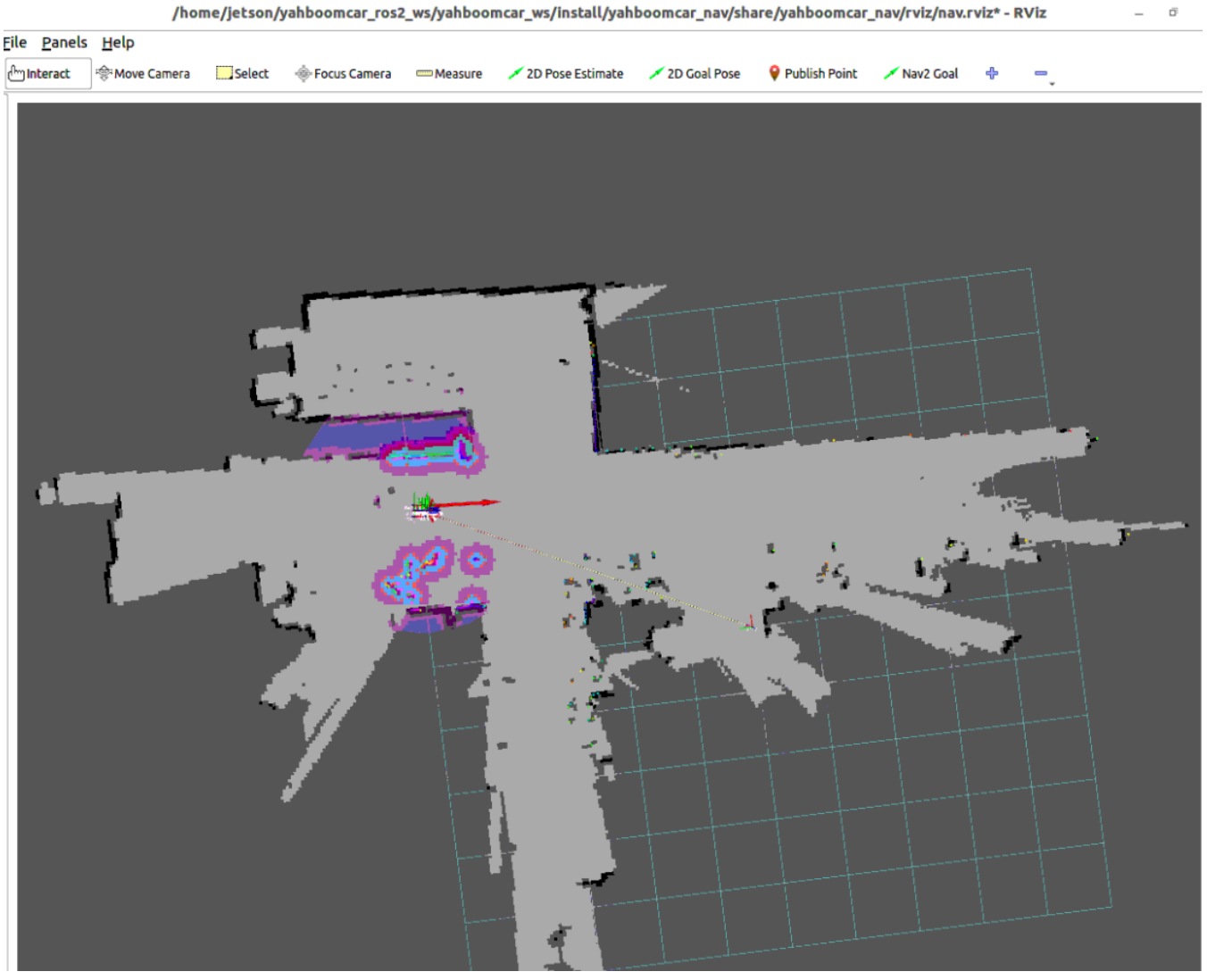


3.4 Start Nav2 Navigation Controller

Run in **Terminal 4** to start navigation planning and controller (loads corresponding static raster map):

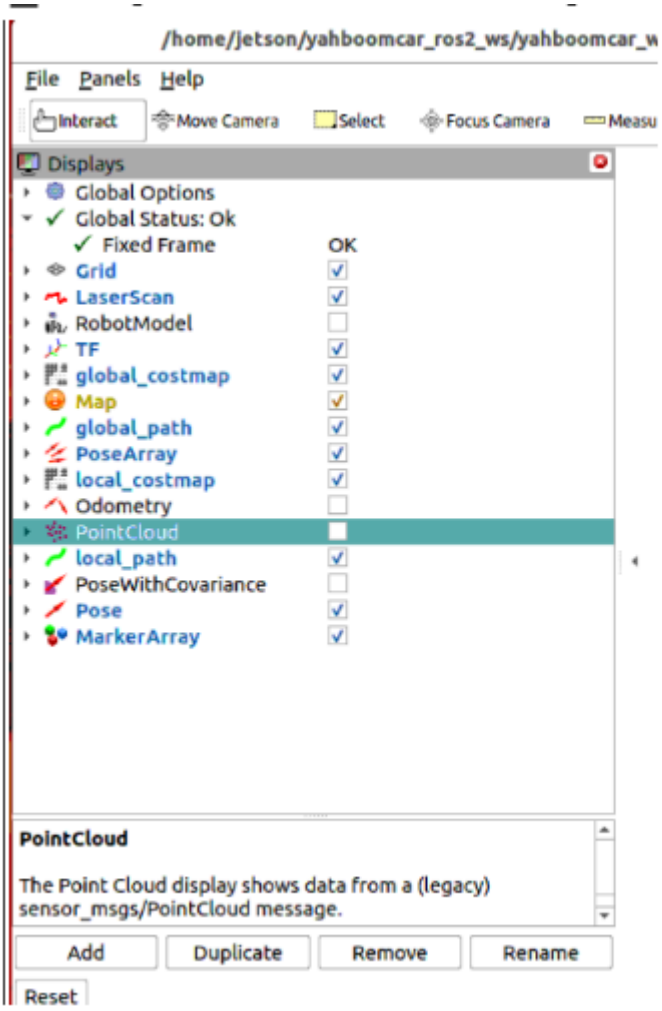
```
ros2 launch yahboomcar_nav navigation_cartodwb_launch.py
maps:=$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/map.yaml
```

```
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws$ ros2 launch yahboomcar_nav navigation_cartodwb_launch.py maps:=/home/jetson/map/map.yaml
[component_container_isolated-1] [INFO] [1788170672.360287403] [behavior_server]: Creating bond (behavior_server) to lifecycle manager.
[component_container_isolated-1] [INFO] [1788170672.467487545] [lifecycle_manager_navigation]: Server behavior_server connected with bond.
[component_container_isolated-1] [INFO] [1788170672.467628992] [lifecycle_manager_navigation]: Activating bt_navigator
[component_container_isolated-1] [INFO] [1788170672.469218311] [bt_navigator]: Activating
[component_container_isolated-1] [INFO] [1788170672.608745067] [bt_navigator]: Creating bond (bt_navigator) to lifecycle manager.
[component_container_isolated-1] [INFO] [1788170672.739241174] [lifecycle_manager_navigation]: Server bt_navigator connected with bond.
[component_container_isolated-1] [INFO] [1788170672.739378300] [lifecycle_manager_navigation]: Activating waypoint_follower
[component_container_isolated-1] [INFO] [1788170672.740677206] [waypoint_follower]: Activating
[component_container_isolated-1] [INFO] [1788170672.740783163] [waypoint_follower]: Creating bond (waypoint_follower) to lifecycle manager.
[component_container_isolated-1] [INFO] [1788170672.847842019] [lifecycle_manager_navigation]: Server waypoint_follower connected with bond.
[component_container_isolated-1] [INFO] [1788170672.847954760] [lifecycle_manager_navigation]: Activating velocity_smoother
[component_container_isolated-1] [INFO] [1788170672.848561411] [velocity_smoother]: Activating
[component_container_isolated-1] [INFO] [1788170672.848706122] [velocity_smoother]: Creating bond (velocity_smoother) to lifecycle manager.
[component_container_isolated-1] [INFO] [1788170672.952698535] [lifecycle_manager_navigation]: Server velocity_smoother connected with bond.
[component_container_isolated-1] [INFO] [1788170672.952832077] [lifecycle_manager_navigation]: Managed nodes are active
[component_container_isolated-1] [INFO] [1788170672.952858926] [lifecycle_manager_navigation]: Creating bond timer...
[component_container_isolated-1] [INFO] [1788170674.245692524] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 281 X 376 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170683.245718747] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 281 X 376 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170688.245735423] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170691.245707579] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170699.245692377] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170708.245719346] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170716.245724174] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170724.245696744] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170732.245724930] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170741.245882878] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170749.245696157] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 278 X 380 at 0.050000 m/pix
```



Tip

In the Views panel on the right side of RViz, you can switch views or adjust observation height:



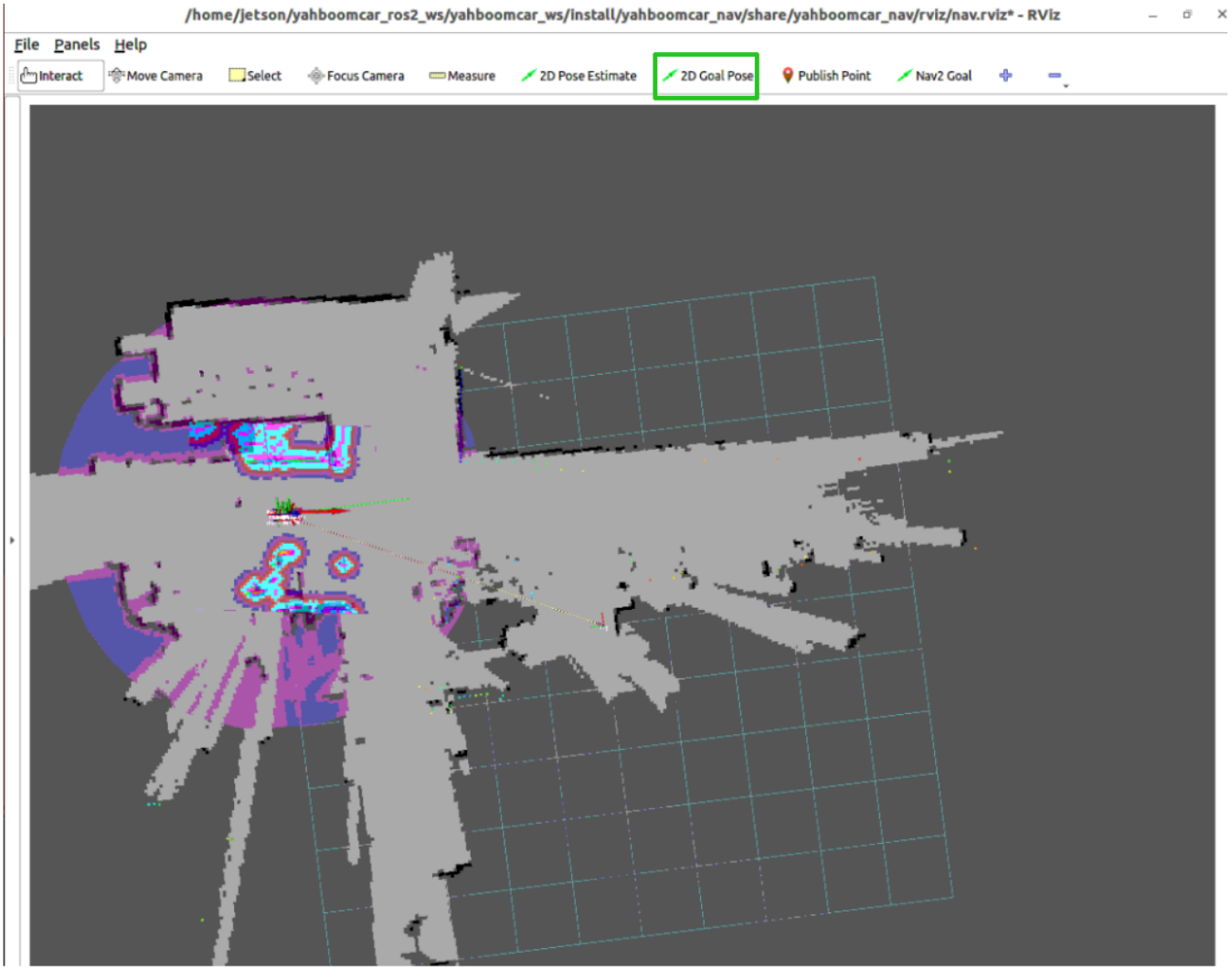
4. Relocalization Navigation Operation Flow and Effect Description

4.1 Automatic Relocalization and Status Check

After Cartographer pure localization starts, the system will automatically match the current laser point cloud with the loaded historical submaps, outputting stable `map -> odom` coordinate transformation. Observe that the laser point cloud in RViz is completely attached to the map contours, representing successful relocalization.

4.2 Single-Point Autonomous Obstacle-Avoiding Navigation

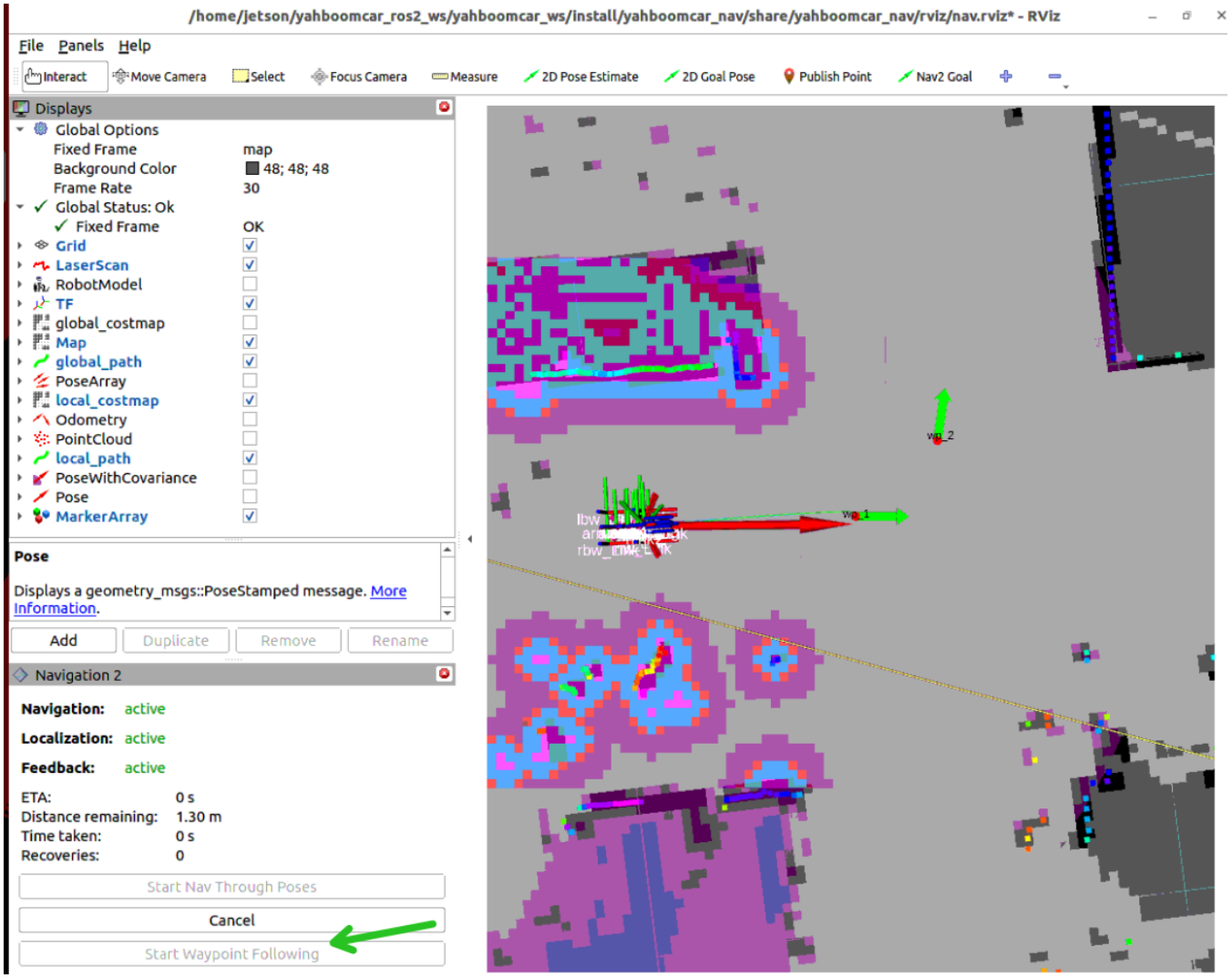
1. Click the `2D Goal Pose` (or Nav2 Goal) tool at the top of RViz2.
2. Click in the free traversable area on the map and drag out the expected goal heading arrow.
3. The system will immediately generate a global planned path, the robot autonomously starts front-wheel steering and smoothly moves forward, real-time avoiding dynamic obstacles during travel and heading to the destination.



4.3 Multi-Point Autonomous Continuous Cruising

If multi-point cruising task is needed:

1. Click `Panels / navigation2 /Waypoint / Nav Through Poses Mode` in the Navigation2 control panel in the lower-left corner of RViz2.
2. Use `Nav2 Goal` to sequentially click and calibrate multiple target waypoints on the map.
3. Click `Start Waypoint Following`, and the robot will sequentially pass through each waypoint in the preset order.



4.4 Goal Arrival and Terminal Feedback

After the robot arrives at the preset goal and completes pose alignment, it automatically stops smoothly, and the terminal will output goal successful completion prompt:

```
borting handle.
[component_container_isolated-1] [INFO] [1788172623.249148862] [local_costmap.local_costmap]: Received request to
clear entirely the local_costmap
[component_container_isolated-1] [INFO] [1788172623.251278541] [controller_server]: Received a goal, begin computi
ng control effort.
[component_container_isolated-1] [WARN] [1788172623.391740876] [controller_server]: Control loop missed its desire
d rate of 20.0000Hz
[component_container_isolated-1] [INFO] [1788172623.996234258] [controller_server]: Passing new path to controller
.
[component_container_isolated-1] [INFO] [1788172625.041833161] [controller_server]: Passing new path to controller
.
[component_container_isolated-1] [INFO] [1788172625.994120859] [controller_server]: Reached the goal!
[component_container_isolated-1] [INFO] [1788172626.033533485] [bt_navigator]: Goal succeeded
[component_container_isolated-1] [INFO] [1788172637.851944607] [global_costmap.global_costmap]: StaticLayer: Resiz
ing costmap to 343 X 456 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788172638.851952912] [global_costmap.global_costmap]: StaticLayer: Resiz
ing costmap to 343 X 457 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788172639.853687307] [global_costmap.global_costmap]: StaticLayer: Resiz
ing costmap to 345 X 458 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788172642.851921741] [global_costmap.global_costmap]: StaticLayer: Resiz
ing costmap to 346 X 448 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788172643.851966339] [global_costmap.global_costmap]: StaticLayer: Resiz
ing costmap to 347 X 448 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788172644.851950517] [global_costmap.global_costmap]: StaticLayer: Resiz
ing costmap to 352 X 453 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788172650.851948323] [global_costmap.global_costmap]: StaticLayer: Resiz
ing costmap to 357 X 453 at 0.050000 m/pix
```

> **Note**
>
> To end navigation task, press `Ctrl+C` in each terminal window to stop the program.

---

5. Core Source Code and Localization Parameter Analysis

5.1 Pure Localization and Navigation Launch Analysis

Pure Localization Launch Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/launch/nav2/localization_imu_odom.launch.py
```

- Loads `cartographer_node`, loads `.pbstream` format pose graph through `-load_state_filename` parameter.
- Starts `cartographer_occupancy_grid_node` to output static raster map layer.
- Starts `set_initpose_from_rviz` node to receive `/initialpose` topic from RViz, achieving interactive relocalization initialization.

Navigation Controller Launch Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/launch/nav2/navigation_cartodwb_launch.py
```

- Introduces `cartographer_bringup_launch.py`, shields AMCL node, with Cartographer providing global localization.
- Loads `cartoteb_nav_params.yaml` navigation parameter file, collaborating Planner and Controller to complete motion planning.

5.2 Core Algorithm Parameter Interpretation

Parameter Configuration File Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/params/cartoteb_nav_params.yaml
```

Core Parameter Description Table

Parameter Module	Parameter Name	Current Config Value	Core Function and Physical Meaning
Cartographer Pure Localization	`load_state_filename`	`yahboomcar.pbstream`	Complete path of loaded offline pose graph state file
Cartographer Optimization	`pure_localization_trimmer`	`max_submaps: 3`	Pure localization trimmer only retains the 3 most recent local submaps, avoiding additional memory usage
Local Controller	`FollowPath.plugin`	`TebLocalPlannerROS`	Local trajectory optimization planner plugin
Local Controller	`max_vel_x`	`0.30`	Maximum allowed cruising speed during navigation (m/s)
Local Controller	`xy_goal_tolerance`	`0.20`	Goal point planar distance arrival tolerance (meters)
Local Controller	`yaw_goal_tolerance`	`0.25`	Goal point heading angle arrival tolerance (radians ≈ 14.3°)
Local Costmap	`inflation_radius`	`0.21`	Local obstacle dynamic inflation radius (meters)
Local Costmap	`cost_scaling_factor`	`8.0`	Scaling factor for obstacle cost decay with distance

Parameter File Core Configuration Snippet

```yaml
# Controller server configuration (TEB Local Planner)
controller_server:
  ros__parameters:
    use_sim_time: False
    controller_frequency: 20.0
    FollowPath:
      plugin: "teb_local_planner::TebLocalPlannerROS"
      max_vel_x: 0.30              # Maximum cruising speed (m/s)
      min_vel_x: -0.05             # Limit reverse speed
      max_vel_theta: 1.0           # Maximum turning angular velocity (rad/s)
      acc_lim_x: 1.0               # Linear acceleration limit
      acc_lim_theta: 1.0           # Angular acceleration limit
```

```yaml
      min_obstacle_dist: 0.1       # Minimum obstacle hard distance (m)
      xy_goal_tolerance: 0.20      # Arrival distance accuracy (m)
      yaw_goal_tolerance: 0.25     # Arrival angle accuracy (rad)

# Local costmap inflation configuration
local_costmap:
  local_costmap:
    ros__parameters:
      inflation_layer:
        plugin: "nav2_costmap_2d::InflationLayer"
        cost_scaling_factor: 8.0
        inflation_radius: 0.21     # Inflation radius (m)
```